

Set up

```
wget https://github.com/junchaoshi/SPORTS1.1/archive/master.zip
```

```
unzip master.zip
```

```
chmod 755 ~/sports1.1-master/source/sports.pl
```

```
export PATH="$PATH:$HOME/sports1.1-master/source"
```

```
echo 'export PATH=$PATH:$HOME/sports1.1-master/source' >> ~/.bashrc  
source ~/.bashrc
```

```
sports.pl -h | head -n 25
```

#install

```
sudo apt-get update  
sudo apt-get install -y sra-toolkit r-base r-base-dev python3-pip  
python3 -m pip install --user --upgrade pip  
python3 -m pip install --user --upgrade cutadapt  
echo 'export PATH=$HOME/.local/bin:$PATH' >> ~/.bashrc  
source ~/.bashrc
```

```
sudo apt-get update  
sudo apt-get install -y bowtie
```

#Demo check

```
cd ~/sports1.1-master/demo  
less Demo_Instruction.md
```

#Download reference genome (mouse)

```
mkdir -p ~/SPORTS_db  
cd ~/SPORTS_db
```

```
wget -O Mus_musculus.tar.gz \  
'https://ncrnainfo-my.sharepoint.com/personal/sports_ncrna_info/_layouts/15/guestaccess.aspx?docid=0833653a140eb47f098267d7a23d3b63c&authkey=Ab8aoYC8paqFI2yRablo7Ok'
```

```
#Homo sapiens (Human)
```

- i. annotation databases: (We provide a download link for all databases listed below:

https://ncrnainfo-my.sharepoint.com/personal/sports_ncrna_info/_layouts/15/guestaccess.aspx?docid=0773ed3d5f6b74f35bbd643e1af221c31&authkey=AcRxf8wainGUIEhgl--8CDc)

```
#extract
```

```
tar -zxvf Mus_musculus.tar.gz
```

Step-by-step: build all required Bowtie indexes (Mouse)

Bowtie1 must be installed

Check first:

```
bowtie --version
```

```
bowtie-build --version
```

1 Genome (required)

```
cd ~/SPORTS_db/Mus_musculus/genome/mm10
```

```
bowtie-build genome.fa genome
```

Result you should see:

```
genome.1.ebwt
```

```
genome.2.ebwt
```

```
genome.3.ebwt
```

```
genome.4.ebwt
```

```
genome.rev.1.ebwt
```

```
genome.rev.2.ebwt
```

2 miRNA (miRBase)

```
cd ~/SPORTS_db/Mus_musculus/miRBase/21
```

```
bowtie-build miRBase_21-mmu.fa miRBase_21-mmu
```

3 rRNA (combine first, then build)

SPORTS expects **one rRNA index prefix**, but you currently have **multiple rRNA FASTAs**.

Combine them:

```
cd ~/SPORTS_db/Mus_musculus/rRNadb
cat mouse_rRNA_*.fa > mouse_rRNA.fa
```

Build index:

```
bowtie-build mouse_rRNA.fa mouse_rRNA
```

4 tRNA (this gives tsRNA)

You have several tRNA FASTAs — SPORTS expects **one prefix**.

Use the mature + mt tRNAs (recommended by SPORTS logic):

```
cd ~/SPORTS_db/Mus_musculus/GtRNadb/mm10
cat mm10-mature-tRNAs.fa mm10-mt_tRNAs_CCA.fa > mm10-tRNAs.fa
```

(If mm10-tRNAs.fa already exists, overwrite is fine.)

Then build:

```
bowtie-build mm10-tRNAs.fa mm10-tRNAs
```

5 piRNA

```
cd ~/SPORTS_db/Mus_musculus/piRBase
bowtie-build piR_mouse.fa piR_mouse
```

6 Ensembl ncRNA (USE GRCm39 instead)

```
cd ~/SPORTS_db/Mus_musculus/Ensembl/release-89
bowtie-build Mus_musculus.GRCm38.ncrna.fa Mus_musculus.GRCm38.ncrna
GRCm39p.*.ebwt
```

7 Rfam ncRNA

```
cd ~/SPORTS_db/Mus_musculus/Rfam/12.3
```

```
bowtie-build Rfam-12.3-mouse.fa Rfam-12.3-mouse
```

#Inspect

```
ls ~/SPORTS_db/Mus_musculus/genome/mm10/genome*.ebwt | wc -l
ls ~/SPORTS_db/Mus_musculus/miRBase/21/miRBase_21-mmu*.ebwt | wc -l
ls ~/SPORTS_db/Mus_musculus/GtRNadb/mm10/mm10-tRNAs*.ebwt | wc -l
ls ~/SPORTS_db/Mus_musculus/rRNadb/mouse_rRNA*.ebwt | wc -l
ls ~/SPORTS_db/Mus_musculus/piRBase/piR_mouse*.ebwt | wc -l
ls ~/SPORTS_db/Mus_musculus/Ensembl/release-89/Mus_musculus.GRCm38.ncrna*.ebwt
| wc -l
ls ~/SPORTS_db/Mus_musculus/Rfam/12.3/Rfam-12.3-mouse*.ebwt | wc -l
```

```
#GRCm39*.ebwt
```

```
ls ~/SPORTS_db/Mus_musculus/Ensembl/release-89/GRCm39p.*.ebwt
```

Run Sports

Your Treatment Samples

```
conda activate bioinfo
cd your_sample_directory
mkdir -p your_sample_directory_path
```

Extract the fastqsanger.gz file into new generated folder LPS_F1 or

*Decompress the 12 FASTQs (keep originals)

```
pigz -d -k *.fastqsanger.gz 2>/dev/null || gzip -d -k *.fastqsanger.gz
```

#Check:

```
ls -l *.fastqsanger | wc -l
```

```
cd your_sample_directory_path
```

```
cp --update=none "your_sample_directory_path"/*.fastqsanger .
```

```
ls -lh
```

Define sample list

```
samples=(  
  "L1.fastqsanger" "L2.fastqsanger" "L3.fastqsanger"  
  "L4.fastqsanger" "L5.fastqsanger"  
  "S1.fastqsanger"  "S2.fastqsanger"  "S3.fastqsanger"  
  "S4.fastqsanger"  "S5.fastqsanger"  
)
```

Set run parameters + output folder

```
sample_dir="./"  
out_root="."  
threads=4
```

Set reference paths (built Bowtie indexes)

```
REF="$HOME/SPORTS_db/Mus_musculus"  
  
GENOME="$REF/genome/mm10/genome"  
MIRNA="$REF/miRBase/21/miRBase_21-mmu"  
RRNA="$REF/rRNadb/mouse_rRNA"  
TRNA="$REF/GtRNadb/mm10/mm10-tRNAs"  
PIRNA="$REF/piRBase/piR_mouse"  
ENSEMBL="$REF/Ensembl/release-89/Mus_musculus.GRCm38.ncrna"  
RFAM="$REF/Rfam/12.3/Rfam-12.3-mouse"
```

Use a hardlink for samples ending in fastq (so SPORTS can read input samples)

```
cd your_sample_directory_path
```

```
for f in *.fastqsanger; do  
  base="${f%.fastqsanger}"  
  rm -f "${base}.fastq"
```

```
In "$f" "${base}.fastq"
done
```

```
#check size non 0 KB
ls -lh
```

Run SPORTS for one sample first

```
sports.pl \
-i "./your_sample.fastq" \
-p 4 \
-M 1 \
-k \
-g "$GENOME" \
-m "$MIRNA" \
-r "$RRNA" \
-t "$TRNA" \
-w "$PIRNA" \
-e "$ENSEMBL" \
-f "$RFAM" \
-o "./your_sample" \
> "./your_sample/run.log" 2>&1
```

#Then:

```
wc -l your_sample/run_this.sh
find your_sample -maxdepth 3 -type f | head -n 50
```

```
find your_sample -type d -name "*_result"
find your_sample -type f -name "*_output.txt" -o -name "*_summary.txt" -o -name
"*_length_distribution.txt" | head -n 50
```

After testing, run for all samples

```
cd your_sample_directory_path
```

```
#replace with the prefixes of .fastq files
samples=(C1 C2 C3 C4 C5)
```

```
for s in "${samples[@]"; do
    echo "======"
    echo "Running SPORTS: $s"
    echo "======"
```

```
# 1) verify input exists and is non-empty
if [[ ! -s "$s.fastq" ]]; then
    echo "ERROR: missing or empty: $s.fastq"
    ls -lh "$s.fastq" "$s.fastqsanger" 2>/dev/null || true
    continue
fi
```

```
# 2) fresh output folder
rm -rf "$s"
mkdir -p "$s"
```

```
# 3) run
sports.pl \
-i "$s.fastq" \
-p 4 \
-M 1 \
-k \
-g "$GENOME" \
-m "$MIRNA" \
-r "$RRNA" \
-t "$TRNA" \
-w "$PIRNA" \
-e "$ENSEMBL" \
-f "$RFAM" \
-o "$s" \
> "$s/run.log" 2>&1
```

```
# 4) quick success check (prints the summary path if generated)
find "$s" -type f -name "$s_summary.txt" | head -n 1
done
```

```
find "$s" -type f -path "**_result/*" | head
```

```
find "$s" -type f -name "${s}_output.txt" -o -name "${s}_summary.txt" -o -name  
"${s}_length_distribution.txt"
```

After batch: collect all summaries into one folder

```
cd your_sample_directory_path  
mkdir -p SPORTS_summaries
```

```
#replace with your file prefixes
```

```
for s in C1 C2 C3 C4 C5; do
```

```
  cp -f "$s/1_$$s/${s}_result/${s}_summary.txt" "SPORTS_summaries/${s}_summary.txt"  
done
```

```
ls -lh SPORTS_summaries
```

Now navigate to the SPORTSDE.R script for differential expression analysis.